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Supplementary Materials for

A CMOS-based highly scalable flexible neural electrode interface

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The PDF file includes:

Figs. S1 to S6
Legend for movie S1

Other Supplementary Material for this manuscript includes the following:

Movie S1

Supplementary Materials

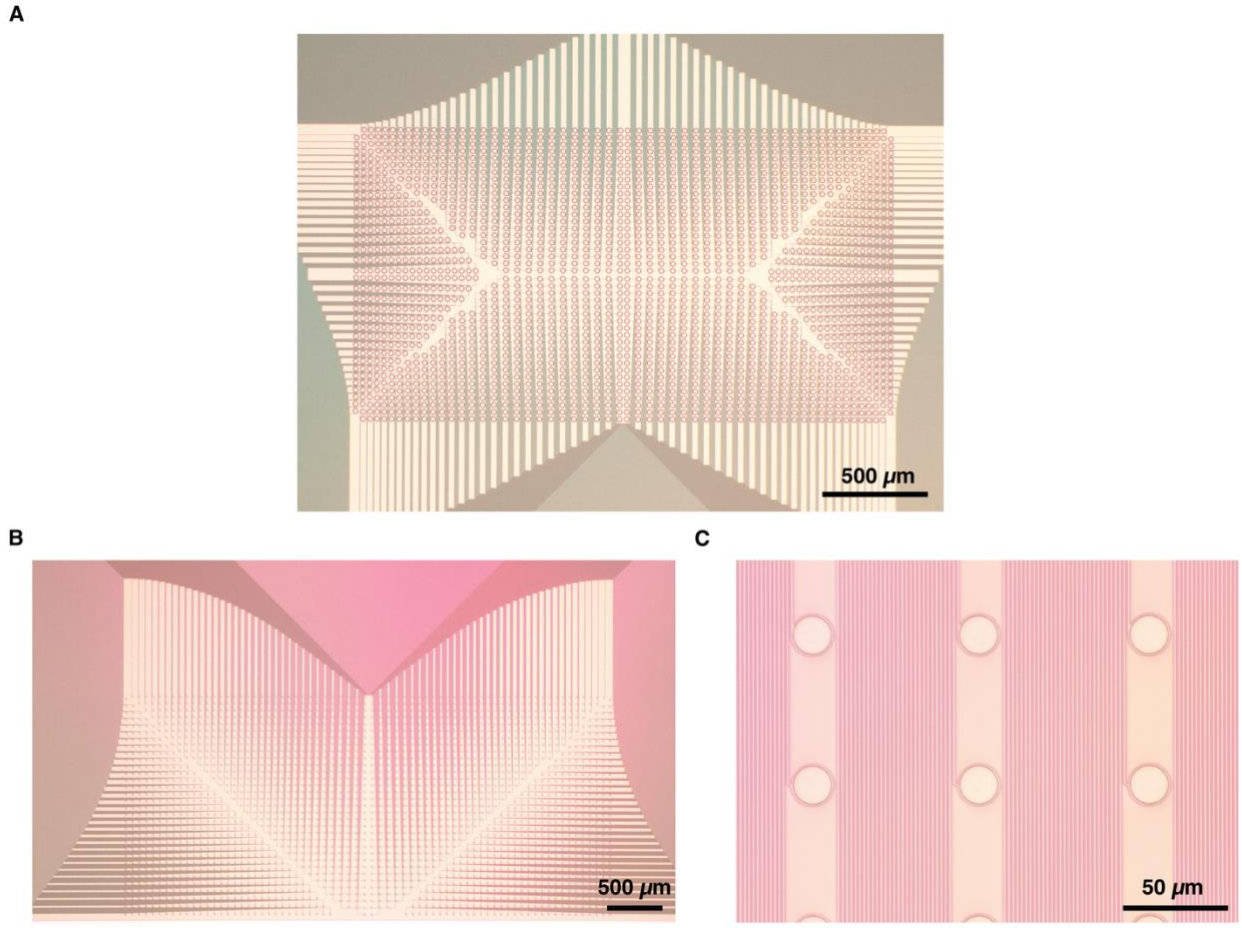


Figure S1. 2200-channel geometry. (**A**) Microscopy image of 2200 Flex2Chip microstructures in a 3.85 mm x 2.10 mm area, with a channel density of 272 channels / mm^2. The routing escapes in all four directions to maximize the density of pads in the active area. (**B**) Microscopy image of the distal end of the 2200-channel device, which interfaces with the neurons. (**C**) An inset of Fig. S1B showing each individual recording/stimulation pad.

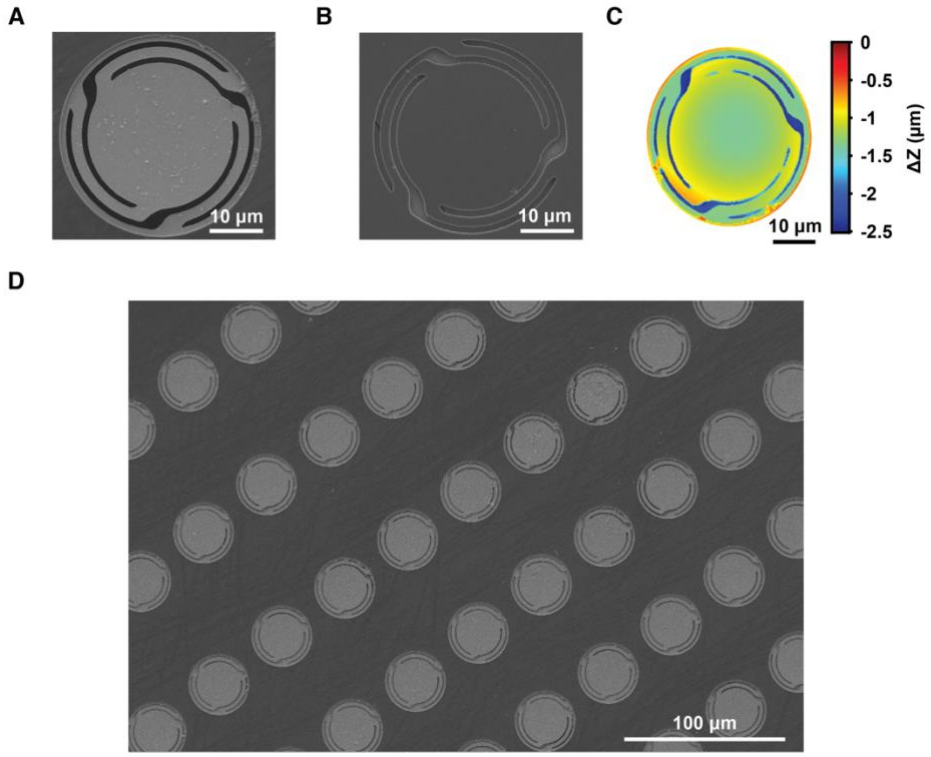


Figure S2. Flex2Chip microstructure characterization. (**A** and **B**) SEM images of the Flex2Chip microstructure, platinum-side facing up and down respectively. Here, the devices were released from the wafer and placed on a glass slide. (**C**) Optical profilometer image of the Flex2Chip microstructure, platinum-side facing up. (**D**) SEM image of an array of Flex2Chip microstructures, platinum-side facing up.

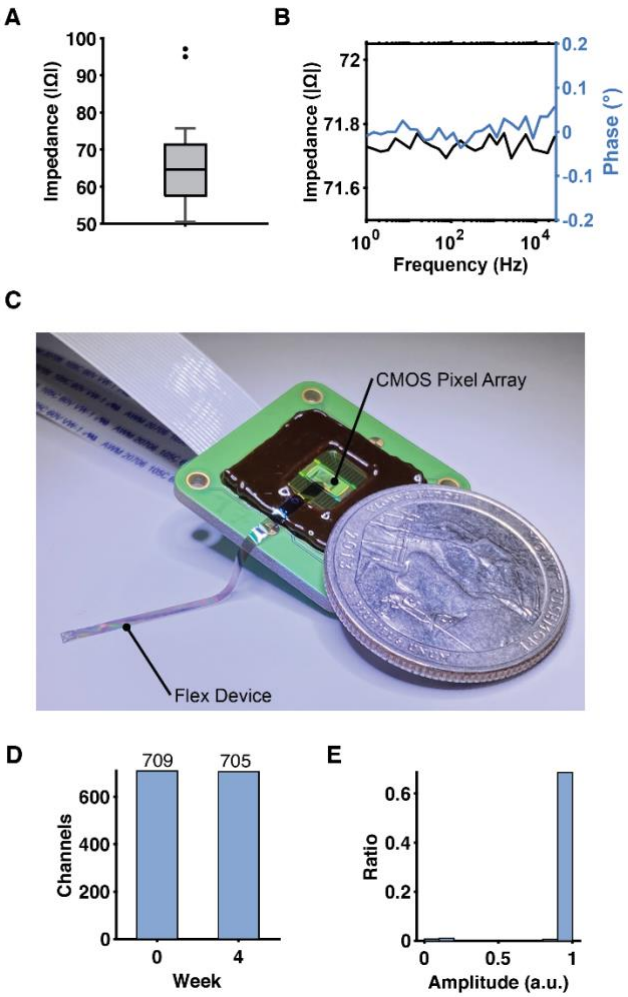


Figure S3. Flex2Chip electrical characterization. (**A**) The resistance of the Flex2Chip interface computed from the IV curves from Fig. 2D. The narrow variation (66.5 ± 13.0 Ω, n = 18) in impedance demonstrates minimal interelectrode variability. (**B**) A representative Bode plot indicates uniform resistivity and phase from 1 Hz to 30 kHz. This finding confirms that there is no capacitive impedance at the physiological frequencies of interest. (**C**) Photograph of an assembled Flex2Chip device on a CMOS-MEA adjacent to a United States quarter. The CMOS-MEA headstage has two 24-channel ZIF connectors leading to a downstream FPGA. (**D**) The connectivity yield is virtually unchanged after leaving it in an incubator at 37 °C and 97% humidity over a period of a month, with a minor decrease of 709 to 705 out of a total of 720 pads. (**E**) The electrical characteristics also remain unchanged, with the aged device detecting 0.99 ± 0.03 of the full-scale signal.

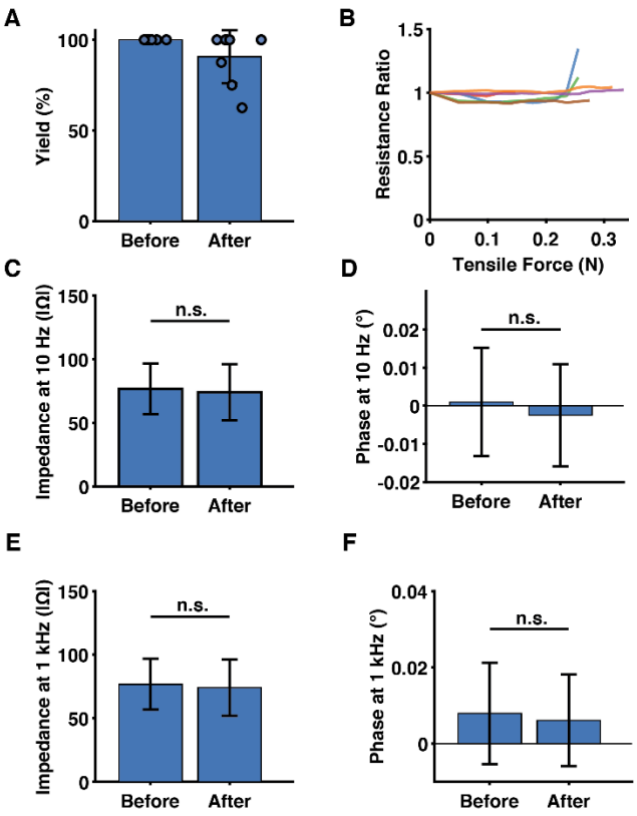


Figure S4. Encapsulation methodology. (A) After assembly, the Flex2Chip interface can be encapsulated with silicone, which seals the microstructures in place and forms a tough mechanically protective layer. The yield of the devices was minimally impacted, with a decrease of 9.38 ± 14.6 % (n = 8 devices, 8 channels per device). **(B)** The robustness of the interface was measured by applying a tensile force across the device until device breakage. Electrical connectivity was maintained up to 268 ± 6.65 mN (n = 6 devices). **(C, D, E** and **F)** There is no statistical difference in the impedance and the phase of the Flex2Chip interface after the silicone encapsulation at 10 and 1000 Hz (n = 8 devices). The changes in impedance and phase are 76.7 ± 20.0 Ω to 74.1 ± 22.1 Ω (p = 0.42), 0.00 ± 0.01 ° to 0.00 ± 0.01 ° (p = 0.21) at 10 Hz and 76.7 ± 20.0 Ω to 74.1 ± 22.2 Ω (p = 0.42), 0.01 ± 0.01 ° to 0.01 ± 0.01 ° (p = 0.51) at 1 kHz.

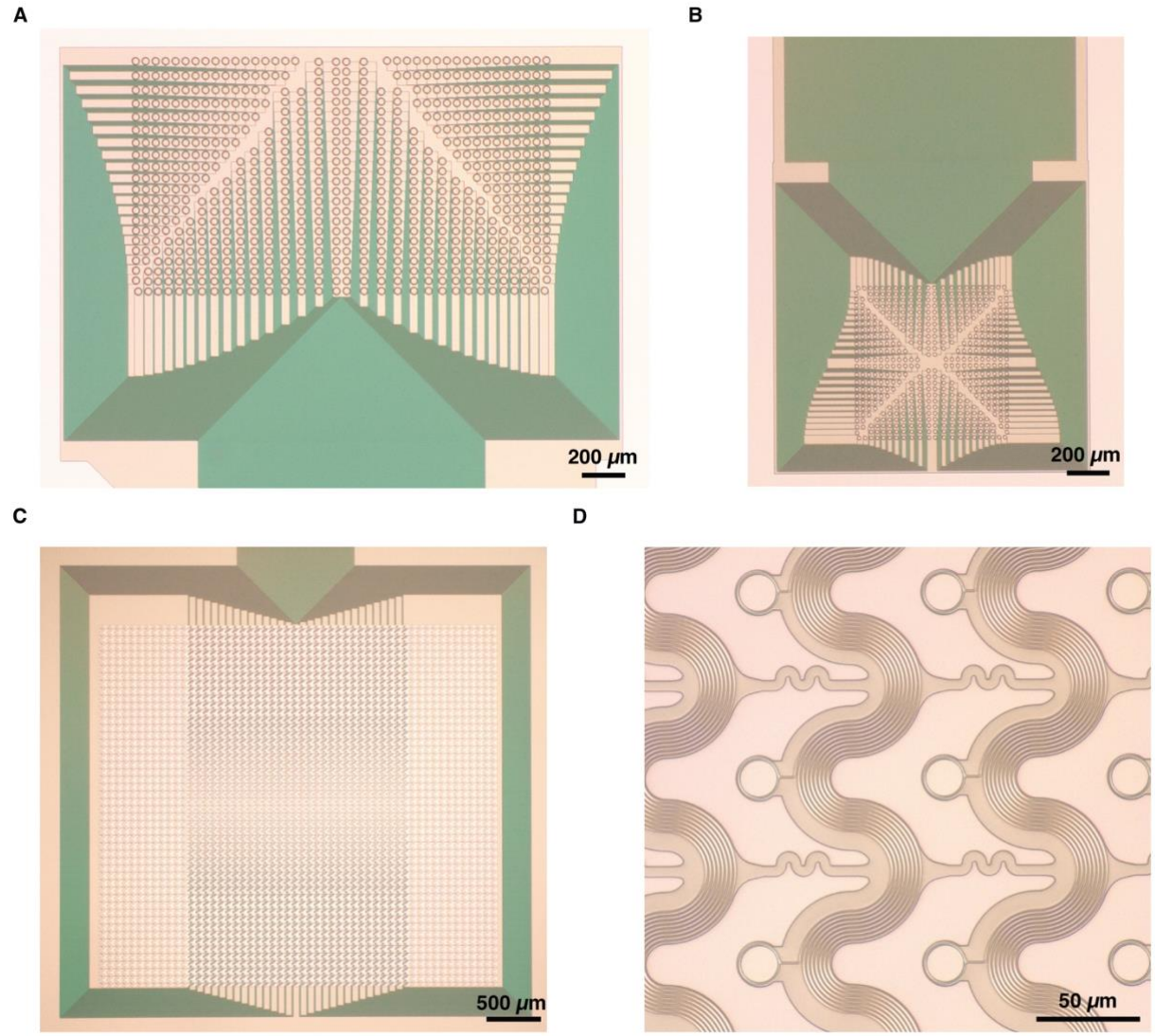


Figure S5. 720-channel geometry. (**A**) Microscopy image of 720 Flex2Chip microstructures in a 2.10 mm x 1.20 mm area, with a channel density of 286 channels / mm^2. This proximal end of the device interfaces with the CMOS-MEA to establish low-noise and Ohmic contact. Each microstructure has an individual 1-µm-wide trace which is routed to the distal end of the device. (**B** and **C**) The distal end of the device can be freely tailored depending on the physiological system in question. The first is an ECoG grid, with 504 20-µm-diameter recording pads in a 0.76 mm x 0.76 mm area, with 216 open channels. The device was designed to fit within a 2-mm-diameter craniotomy for awake head-fixed recordings. The second is a fenestrated grid of 720 20-µm-diameter recording pads in a 2.63 mm x 2.09 mm area. This design was used for brain slice experiments where the fenestrations allow nutrient and oxygen diffusion in the slice during the recording sessions. (**D**) An inset of Fig. S5C showing the individual recording pads.

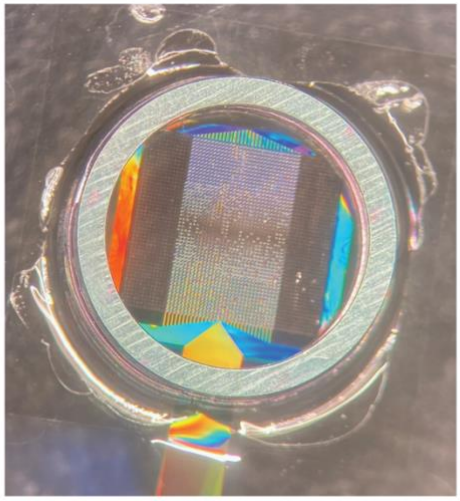


Figure S6. In vitro device setup. The 720-channel device has an active area of 2.63 mm x 2.09 mm with electrodes at a 90 µm pitch. A 6-mm-diameter stainless-steel ring was attached to the device using superglue so that sufficient mechanical contact was made when placed on top of the slice. The fenestrations allow for nutrient and oxygen diffusion so that the slice remains healthy for the duration of the experiment.

Movie S1. Seizure propagation patterns in absence epilepsy. Seizure propagation patterns across a 0.76 mm x 0.76 mm area over the exposed motor and somatosensory cortex. (Left) Heatmap of the generalized phase with black contour lines. (Right) Recorded voltage traces, color coded by the generalized phase. Seizure propagation trajectories can be observed to be highly dynamic across a single SWD: left at 2.186 s, top left at 2.210 s, top right at 2.235 s. Distinct functional boundaries can also be observed at 2.044 s, where the top half is stationary, and the bottom half is propagating to the left. Individual SWDs have distinct propagation patterns indicating a different site of onset.